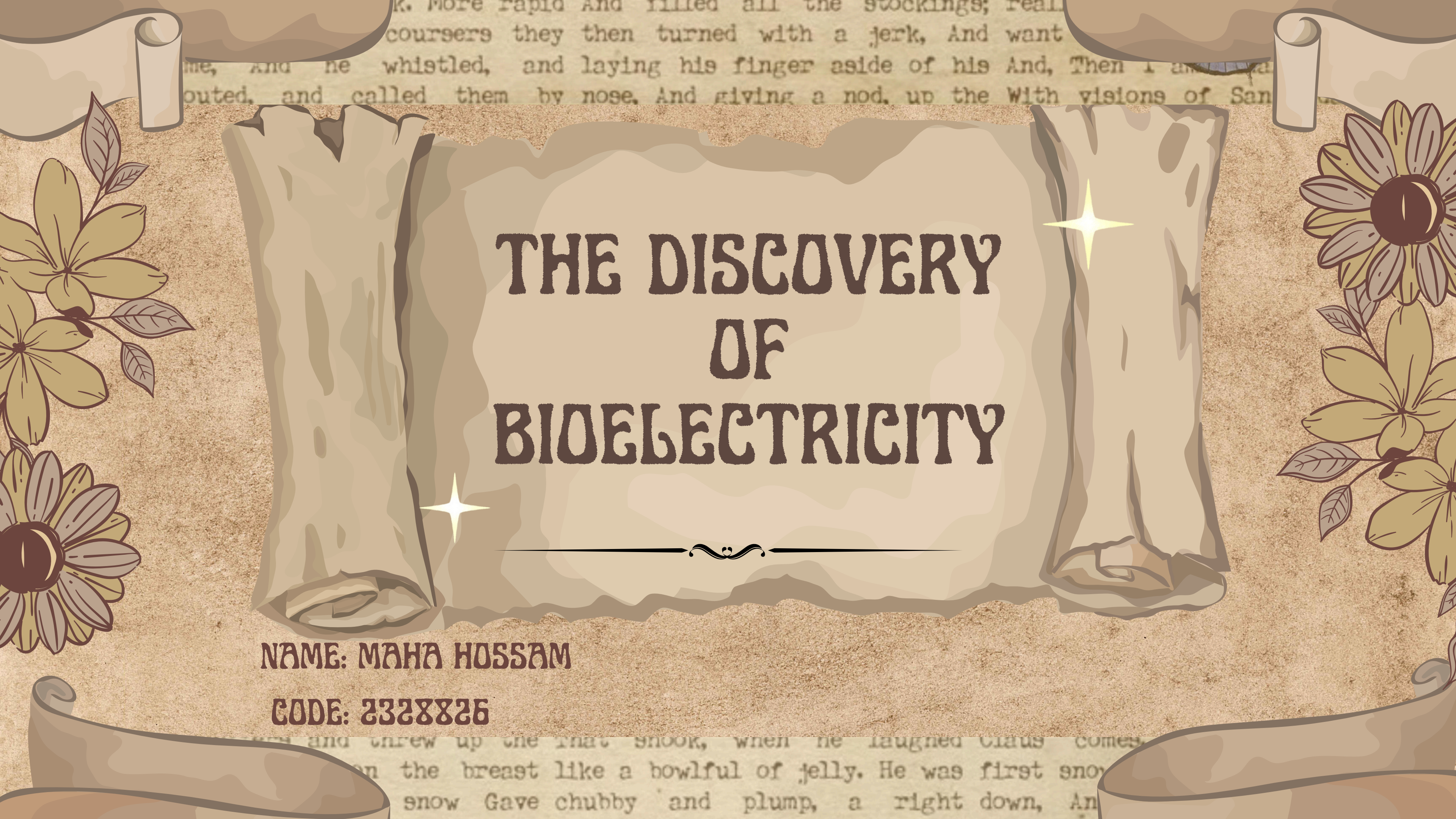




THE DISCOVERY OF BIOELECTRICITY

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EARLY DISCOVERIES

The critical events of this history took place in the decade back to BC.

Ancient Egyptians used electric eels for medical purposes believing in their healing properties.

Also, they used the electric fish in electrotherapy to treat some conditions such as joint pain, migraines, and epilepsy.



GALVANI'S

EXPERIMENTS

Galvani noticed something surprising while working with frog legs in the late of 18th century. When his assistant accidentally touched a frog's nerve with a metal instrument, the legs twitched as if they were alive.

Galvani believed this reaction was due to an animal electricity within the frog's body which could be released or activated by certain conditions.





VOLTA'S

EXPLANATION

"Volta accepted Galvani's findings but explained it in another way.

He believed they were caused by the interaction of two different metals in contact with the frog's tissue. He eventually created the first battery, known as the Voltaic Pile, which produced a continuous electric current.



BIOELECTRICITY IN PLANTS

During the Victorian era, scientists began exploring similarities between animal and plant physiology, with a deep focus on electricity within their systems.

Among these scientists were Darwin and Sanderson, who discussed that plants might have electrical properties similar to those in animal muscles and nerves.

General Secretary, Mr. Etheridge, Mr. Merrifield, Mr. Newmarch, and Prof. Ramsay, having been nominated by the President, were elected to be the ballot Auditors of the Treasurer's Accounts on the part of the Society. The Presents received were laid on the Table, and thanks ordered for them.

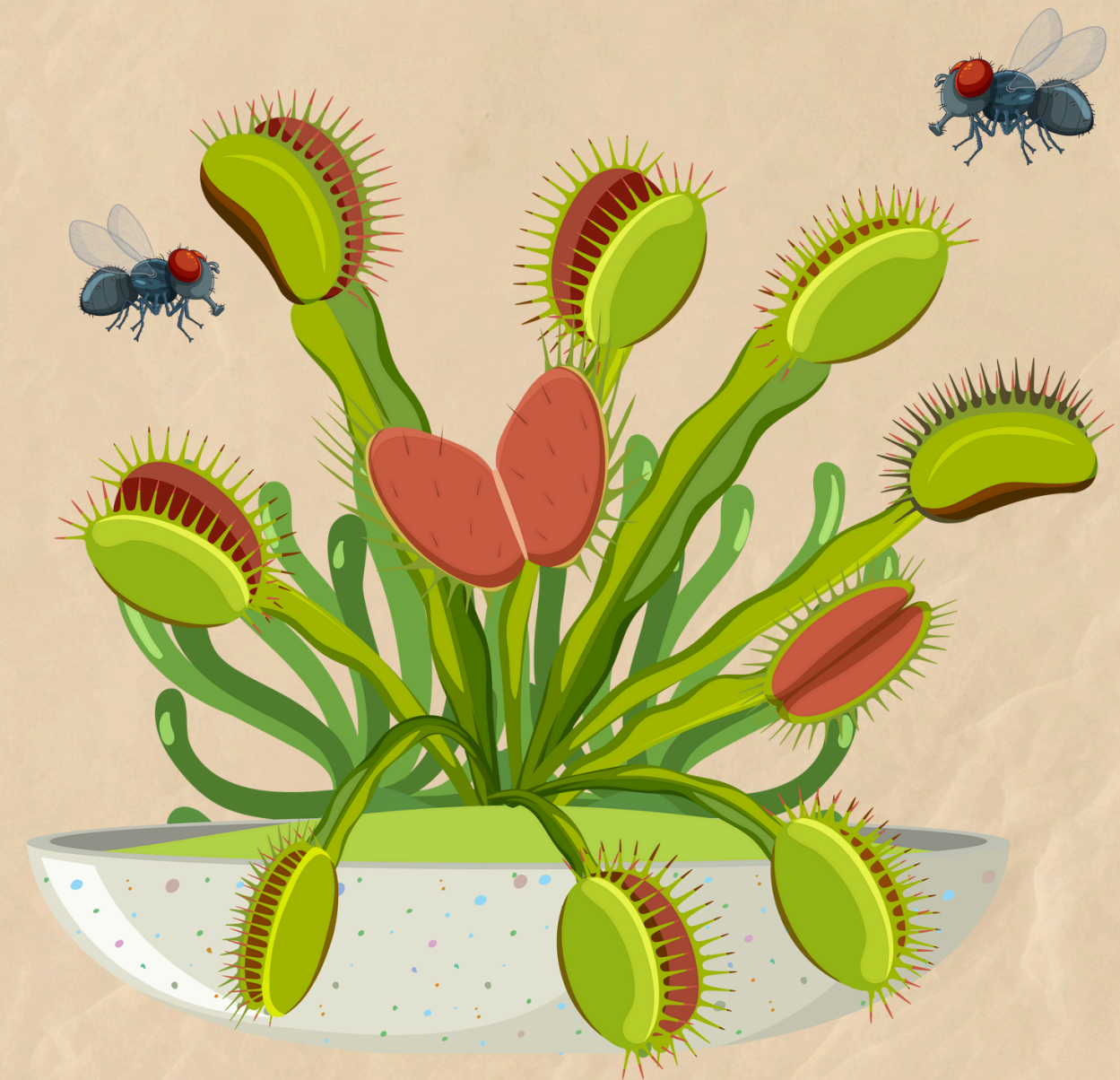
The following communications were read:—

L. "Note on the Electrical Phenomena which accompany irritation of the leaf of *Dionaea muscipula*." By J. BURDON SANDERSON, M.D., F.R.S., Professor of Practical Physiology in University College. Received October 13, 1873.

1. When the opposite ends of a living leaf of *Dionaea* are placed on non-polarizable electrodes in metallic connexion with each other, and a Thomson's reflecting galvanometer of high resistance is introduced into the circuit thus formed, a deflection is observed which indicates the existence of a current from the proximal to the distal end of the leaf.

In 1873, Sanderson conducted the first experiment on the Venus flytrap, discovering that when the fly touches their leaves there are electrical signals produced.

Plants, like animals, have electrical mechanisms that aid in various functions.



CURRENTLY RESEARCHES

Scientists study how electrical signals help control ion transport from the soil, in addition to how plants open and close tiny pores (stomata) on their leaves for gas exchange.

They found that the ion channels in plants are similar to those in the animal nervous system, which help them to control the defense signals in response to damage or attack.





THANKS